

Sujay Shankar

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🌐 github.com/Sujay-Shankar

Education

The University of Texas at Austin

Bachelor of Science: Astronomy

Bachelor of Science: Computational Physics

Certificate: Elements of Computing

August 2020 – December 2023

GPA: 3.8927

GPA: 3.9058

GPA: 4.0000

Boston University

Doctor of Philosophy: Astronomy (in progress)

September 2024 – present

GPA: 3.93

Positions

Boston University | Center for Space Physics

Graduate Research/Teaching/Dean's Fellow

September 2024 – present

Boston, MA

- As a Teaching Fellow, assisted with instruction and grading for *AS 202: Principles of Astronomy I*
- Ran SWMF simulations of magnetic reconnection in solar flares with magnetohydrodynamics and particle-in-cell methods
- Developed algorithms for post-processing and visualizing SWMF simulation data
- Photometrically and polarimetrically observed UV Ceti variables (flare stars) with PRISM aboard the Perkins telescope
- Acted as a Department of Astronomy Graduate Student Representative for the 2025-2026 academic year

The University of Texas at Austin | Department of Astronomy

Research Engineering/Scientist Assistant

January 2024 – July 2024

Austin, TX

- Added `gollum` support for newly released Sonora Diamondback brown dwarf atmospheric models
- Improved `gollum`'s documentation, setup, testing, and directory management systems
- Submitted papers for `blase3D` to ApJ and `gollum` to JOSS

The University of Texas at Austin | Department of Astronomy

AST 375C: Conference Course in Astronomy

August 2023 – December 2023

Austin, TX

- Lead developer of `blase3D`, a fork of `blase` (Gully-Santiago & Morley 2022)
- Used interpretable machine learning with GPUs to clone PHOENIX spectra across T_{eff} , $\log(g)$, and $[\text{Fe}/\text{H}]$
- Used linear interpolators to create manifolds mapping stellar properties to line-by-line properties

University of Florida | Department of Astronomy

REU Student Researcher

May 2023 – August 2023

Gainesville, FL

- Synthesized a globular cluster escapee sample from APOGEE DR17 and GALAH DR3, combined with Gaia dynamics
- Developed a multithreaded orbit integration pipeline with Monte Carlo initial conditions
- Used chemical, dynamical, and photometric information to match escapee candidates with globular clusters

The University of Texas at Austin | McDonald Observatory

Undergraduate Research Assistant

May 2022 – August 2023

Austin, TX

- Lead developer of the `gollum` Python library, analyzing and visualizing stellar and substellar atmosphere models
- Software architecture improvements, UI/UX improvements, and bug fixes
- Added support for starspot two-component mixture modeling with PHOENIX
- Tested functionality on IGRINS spectra

Publications

- Shankar, S. et al. 2024 *gollum: An intuitive programmatic and visual interface for precomputed synthetic spectral model grids*. JOSS
- Shankar, S. & Gully-Santiago, M. & Morley, C. 2025 *A New Hybrid Machine Learning Method for Stellar Parameter Inference*. ApJ
- Shankar, S. & Bandyopadhyay, A. & Ezzeddine, R. 2025 (in prep) *Tagging Globular Cluster Escapee Candidates from APOGEE and GALAH*. ApJ
- Shankar, S. (in prep) *hydron: A Python interface for launching and post-processing SWMF magnetohydrodynamic simulations*. JOSS
- Shankar, S. et al. 2025 (in prep) *Modeling Fast Magnetic Reconnection in Solar Flare Current Sheets*. ApJ
- Lu, H.-P. et al. 2026 (in prep) *Stellar Flare Observations of AD Leonis with the Perkins and Lowell Discovery Telescopes*.

Technical Skills

Languages: Python, Bash, $\text{\LaTeX}$, MATLAB

Frameworks: Pandas, Xarray, Altair, PyTorch, SWMF

Contexts: Data Analysis/Visualization, High-Performance Computing (TACC, HiPerGator, Perlmutter, Pleiades), Software Packaging, Multi-Platform Development

Conferences

American Geophysical Union 2025 (AGU25) (*upcoming*)

– Presentation: *Modeling Fast Magnetic Reconnection in Solar Flare Current Sheets*

December 2025

New Orleans, LA

2025 Joint CEDAR/GEM Workshop

June 2025

– Poster: *Modeling Magnetic Reconnection in Ionospheric Ejection Current Sheets*

Des Moines, IA

American Astronomical Society 243rd Meeting

January 2024

– Poster: *Novel Dynamical Tagging of Globular Cluster Escapee Candidates back to their Sources*

New Orleans, LA

2023 Bash Symposium

October 2023

– Poster: *Precision Fundamental Stellar Properties with Interpetable Machine Learning*

Austin, TX

Fellowships & Awards

The University of Texas at Austin | Department of Astronomy Board of Visitors Scholarship

2023

Boston University | Dean's Fellowship

2025

2025 Joint CEDAR/GEM Workshop | Travel Support Award

2025